Project Title
Clinical Record Linkage & Care-Coordination Platform

Data Structures and Algorithms -3 (25CS2103E)

in

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ABSTRACT

The Clinical Record Linkage & Care-Coordination Platform addresses the persistent problem of fragmented patient data across hospitals, clinics, diagnostic labs, and pharmacies. In the absence of a universal patient identifier, the same individual's records are scattered across systems that cannot recognise one another.

This project designs and evaluates a probabilistic record-linkage engine that ingests HL7 FHIR-formatted records, resolves duplicate patient identities using a blended deterministic and machine-learning matching approach, and builds a longitudinal Master Patient Index (MPI). A care-coordination dashboard then surfaces the unified record to clinicians, flags conflicting medications, and alerts care teams to gaps in follow-up care.

On a synthetic multi-provider dataset of 250,000 patient records, the system achieved a linkage accuracy of 96.8% with a false-positive rate below 0.4%, demonstrating that entity resolution can be both fast and safe enough for clinical use.

Signature of the Guide